



Visualizing the No-Cloning Theorem

Team Name: Observers

Team Members:

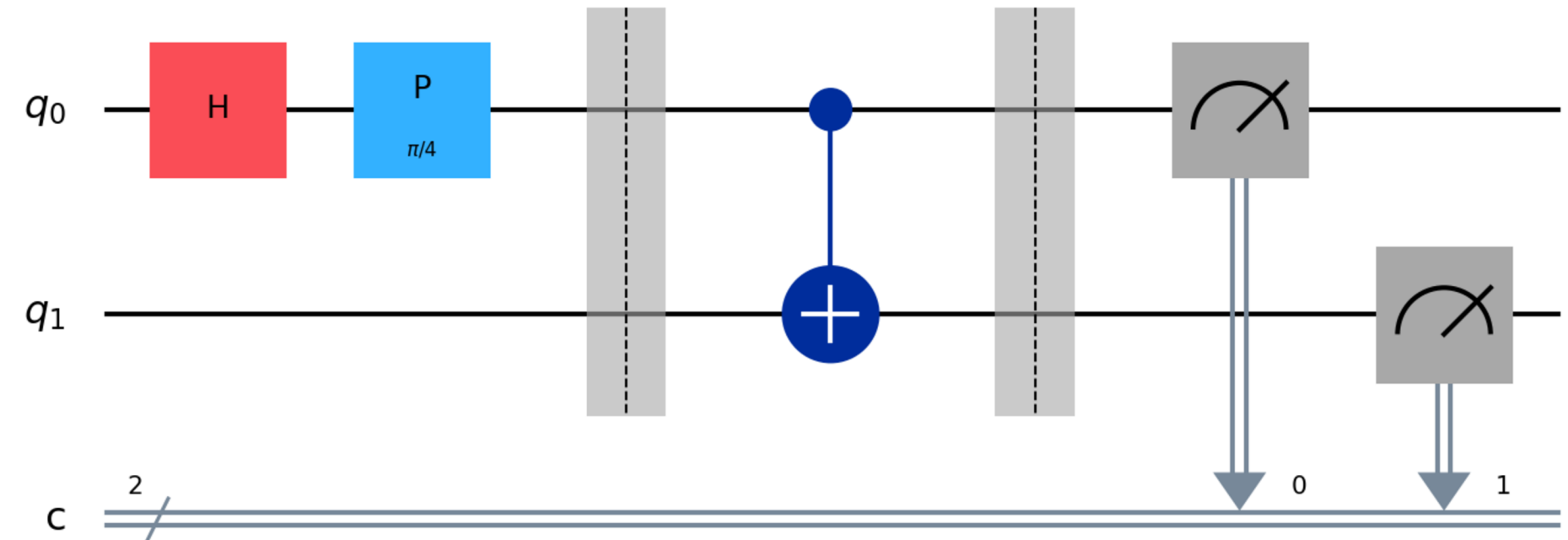
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Problem Identification



- **The Challenge:** In classical networks, data can be copied and intercepted without leaving any physical trace or alert.
- **The Quantum Edge:** We utilize the No-Cloning Theorem, which dictates that an "unknown" quantum state cannot be copied without being physically altered
- **Goal:** To build a self-auditing communication channel that automatically detects eavesdropping

Project Scope & Constraints

Key Objectives and Limitations

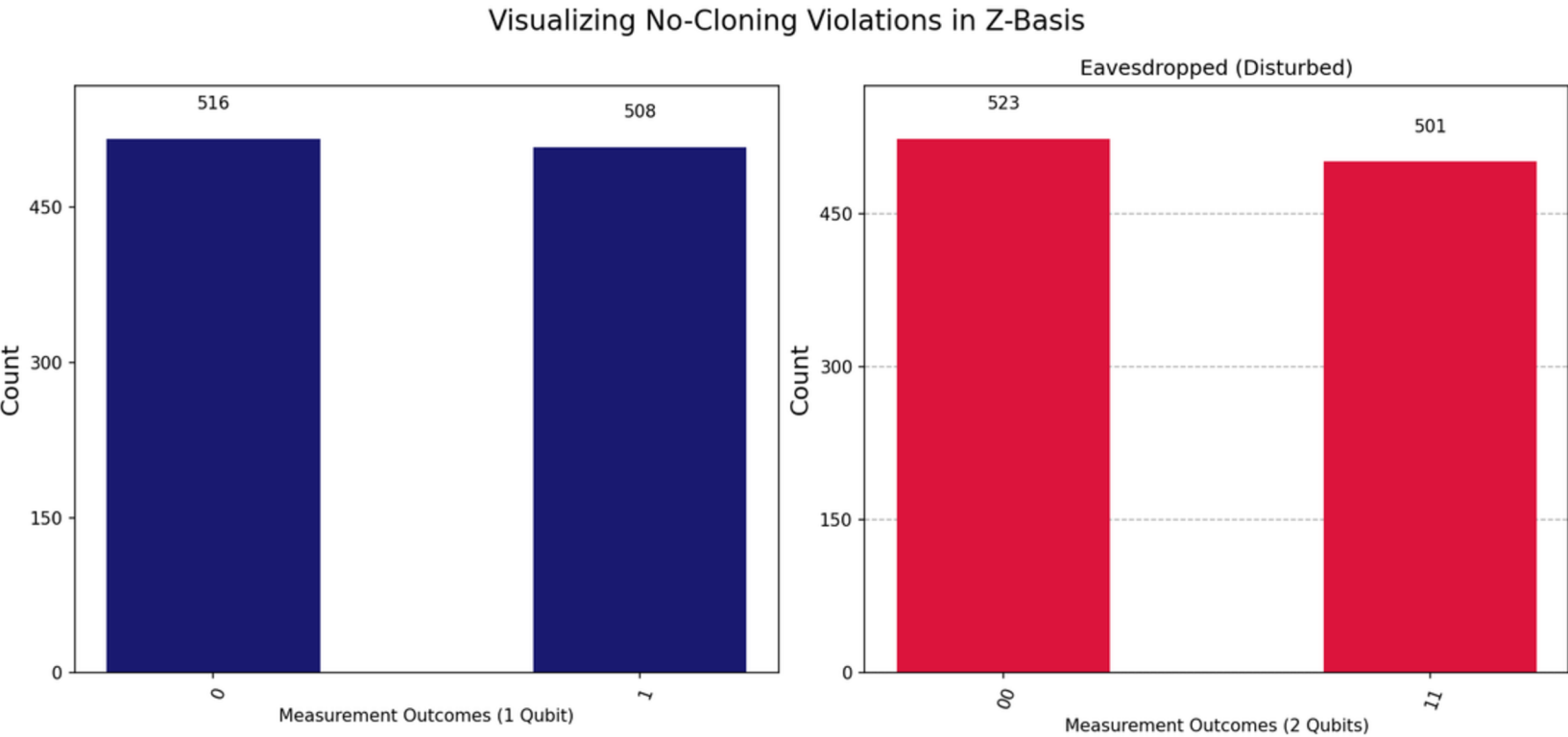
- **Objective:** Prove cloning is impossible
- **Mechanism:** Eve's CNOT interaction
- **Success Metric:** Detection occurs when the error rate exceeds the industry-standard 11% security threshold.
- **Constraints:** The message must be prepared in a non-orthogonal state (using H + P gates) so it remains "unknown" to the attacker.
- Entanglement creates detectable noise



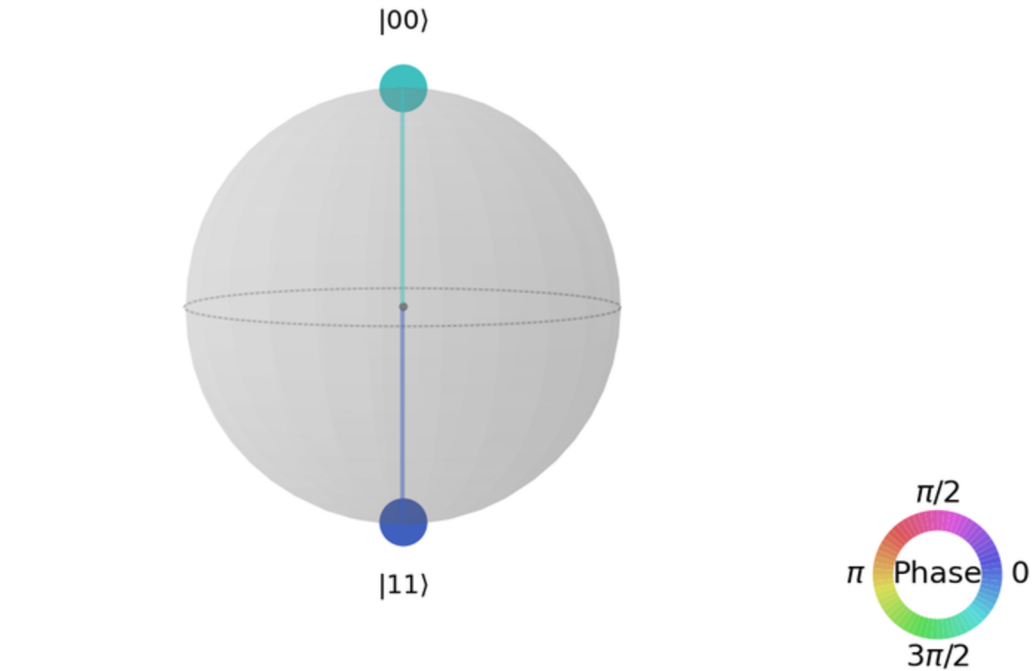
Implementation Strategy

- **Framework:** Built with Qiskit using the BasicProvider for consistent, cross-platform simulation.
- **The Protocol:**
 - Alice: Encodes a message using a Phase (P) gate at $\pi/4$.
 - Eve: Attempts to clone the message using a CNOT gate, which unintentionally creates Entanglement instead of a copy.
- **Security Barriers:** Used `qc.barrier()` to prevent simulator optimization, ensuring Eve's attack represents a realistic physical interception.

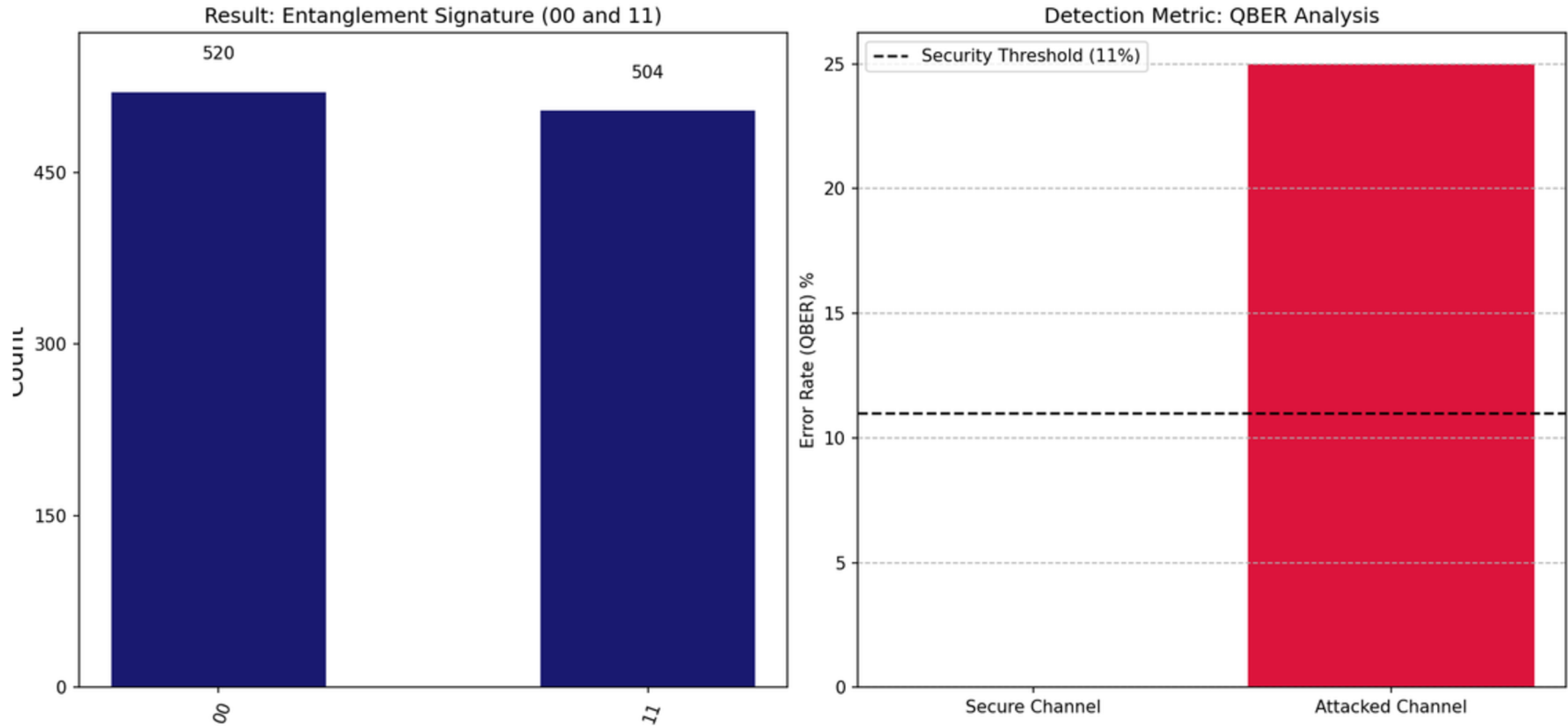
OUTPUTS



Graph 1: Comparative Histogram (Statistical Proof)



Graph 2: QSphere (Global Mapping)



Graph 3: QBER Analysis

Results and Analysis

- Our "Attacked Channel" generated a error rate more than double the 11% Security Threshold.
- Conclusion: The eavesdropping attempt is mathematically and physically impossible to hide.

The Entanglement Trap: Eve's attempt to copy the state via CNOT forced Alice's qubit into a Bell-state entanglement.

Practical Feasibility: Proves that quantum-secured channels (like those needed for Banking or Aadhaar operations) are self-auditing.

Current Limitation: While 100% accurate in a simulator, real hardware noise would require Error Mitigation to distinguish "Eve" from "Decoherence"